

IND-DSP: Improving Night Domain generalization segmentation via Diffusion Semantic Prior

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Abstract—This work addresses the challenge of domain generalization (DG) in nighttime semantic segmentation, which is crucial for autonomous driving. Deep Neural Networks (DNNs) often fail to generalize across domains, particularly under low-light conditions. We propose IND-DSP, an innovative approach leveraging Diffusion Semantic Prior (DSP) techniques combined with Diffusion Denoising Probabilistic (DDP) and Iterative α -(de)Blending (IADB) models. Our method enhances the structural awareness of DG models, improving performance without requiring retraining with synthetic data. Using synthetic nighttime images generated by DGInStyle for training, IND-DSP integrates as a plug-and-play module to existing DG frameworks, enhancing segmentation accuracy. Validated with the GTA source trained RobustNet model on the DarkZurich dataset, our approach improves mean Intersection-over-Union (mIoU) from 7.3% to 18.16%, outperforming the state-of-the-art method. This research highlights the significance of diffusion models as semantic prior for enhancing DG model robustness and offering a computationally efficient solution for domain discrepancies in real-world applications.

I. INTRODUCTION

When deploying deep neural networks (DNNs) trained on a specific dataset (source domain) to real-world unobserved data (target domain), DNNs frequently under perform owing to domain shift. Addressing this issue is essential, particularly for safety-critical applications like autonomous driving. Specifically, real-world data comprises unforeseen and novel examples, such as photos captured in varying illumination, inclement weather conditions, or from disparate locales. Modeling a comprehensive data distribution with minimal training data is typically unfeasible; thus, minimizing the domain

disparity between source and target domains has been a persistent challenge in computer vision. Domain Generalization (DG) is a machine learning technique aimed at improving model performance on unseen domains without access to target domain data during training. Key strategies in DG include learning domain-invariant features, which capture the underlying content rather than superficial characteristics specific to a domain. DG is particularly valuable in fields like autonomous driving and medical imaging, where the cost or feasibility of collecting comprehensive training data is limited. RobustNet [1] is a Domain Generalization Segmentation model that proposes an instance selective whitening (ISW) loss that disentangles feature representations into domain-specific style and domain-invariant content. By selectively removing style-sensitive covariances while retaining content-related features, the method improves robustness against variations in illumination, weather, and other unseen scenarios. Unlike traditional whitening, which removes all feature correlations, ISW targets only those that cause domain shifts, reducing computational overhead.

[3] indicates that a higher incidence of fatal accidents transpires in unfavorable weather conditions, such as nighttime, despite reduced traffic volume during those hours compared to daylight. Consequently, enhancing nighttime segmentation efficacy holds significant importance for autonomous driving contexts. However, RobustNet [1] trained on the GTA dataset performs subpar on the DarkZurich (a nighttime semantic segmentation dataset), indicating it needs more structural prior to urban driving scenes, it is evident from Figure 1 and its qualitative performance from Figure 2.

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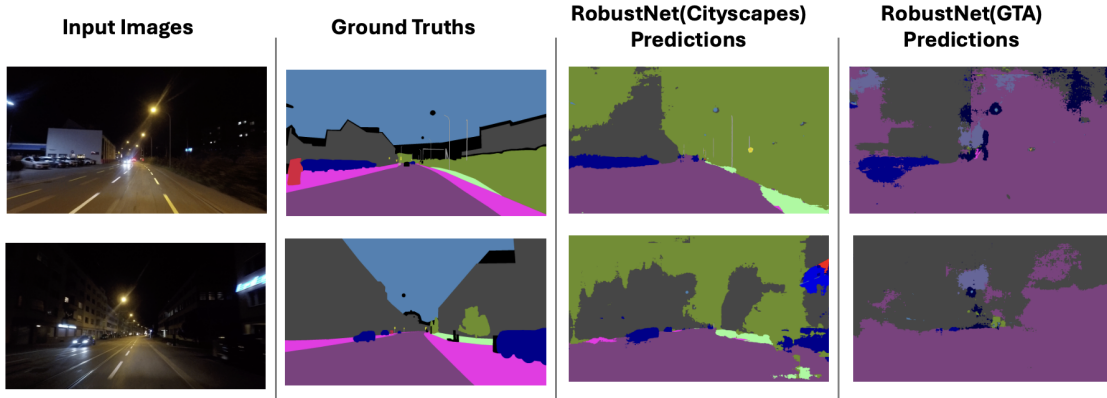


Fig. 1: RobustNet DG model [1] Sub-par Qualitative Performance on DarkZurich [2]

Class	RobustNet(Cityscapes) mIoU%	RobustNet(GTA) mIoU%
0,Road	81.1	32.3
1,Sidewalk	32.0	2.5
2,Building	52.5	36.1
3,wall	25.2	11.4
4,fence	11.8	0.3
5,Pole	19.5	0.3
6,Traffic light	16.7	6.7
7,Traffic sign	9.9	0.0
8,Vegetation	41.5	11.1
9,Terrain	10.9	2.1
10,Sky	0.1	0.0
11,Person	19.0	3.6
12,Rider	8.6	0.0
13,Car	54.7	25.4
14,Truck	0.0	0.0
15,Bus	0.0	0.00
16,Train	0.0	0.0
17,Motorcycle	10.7	0.9
18,Bicycle	15.1	0.0
mIoU	21.53%	7.3%

Fig. 2: RobustNet DG model [1] Sub-par Quantitative Performance on DarkZurich [2]

One of the recent works, DGInStyle [4], which is the state-of-the-art for improving domain generalization (DG) models for semantic segmentation, proposes a novel framework that leverages Latent diffusion models (LDMs) to accomplish its task, particularly in autonomous driving. DGInStyle addresses this by generating diverse, synthetic training data with varied styles, enhancing model robustness against domain

variations like weather, lighting, or geographical differences. One of the drawbacks of this approach is that after generating the synthetic images, the domain generalization model is required to retrain from scratch using the source data, and the generated synthetic data requires extra computation costs. Our method circumvents this issue and also further improves mIoU performance without naively using source and generated DG-InStyle data for retraining the DG model.

Our contributions to this field are three-fold:

- 1) We propose novel training and sampling strategy utilizing DDP and IADB techniques to reconcile the discrepancies between erroneous DG predictions and the actual ground truth. We trained our model using the synthetic images acquired by the DGInStyle technique.
- 2) Our suggested model functions as a plug-and-play module, seamlessly integrating into current DG models and substantially enhancing generalization capability with minimal computational expense.
- 3) We validate the performance of our strategy using the RobustNet model (Source trained on GTA), where our method significantly improves the mIoU performance from the original 7.3% to **18.16%**. In contrast, the

present state-of-the-art DGInStyle technique achieves an improvement of 12.73%.

II. MATERIALS AND METHODS

A. Data Description

RobustNet [1] domain generalization model was trained using the GTA V [5] dataset as its source domain. GTA V is a substantial collection including 24,966 driving-scene images (1914×1052 resolution) created by the Grand Theft Auto V game engine. We utilized DGInStyle-generated nighttime images [4] from the GTA source dataset to train our IND-DSP model, which consists of 1000 synthetic nighttime images (512×512 resolution) derived from urban locations in the GTA V dataset using the DGInStyle proposed data augmentation technique. We used the DarkZurich [2] dataset to evaluate our IND-DSP model; it is an urban road nighttime semantic segmentation image dataset comprising 50 high-resolution (e.g., 1980×1080) annotated nighttime scenes in its validation set.

B. Diffusion models for Segmentation and Data Density Translation

Diffusion models [6] are generative frameworks that transform random noise into structured data through an iterative denoising process. They work by progressively adding noise to training data and then learning to reverse this process. At inference, the model starts with pure noise and iteratively refines it to generate realistic outputs, such as images or audio. Diffusion models have gained popularity due to their high-quality results and Sample diversity, often outperforming traditional generative adversarial networks (GANs). Their flexibility in using conditions along the input and denoising in the latent space [7] make them a powerful tool in modern generative AI.

1) *DDP*: DDP (Diffusion Model for Dense Prediction) [8] is a framework that modifies continuous time and space diffusion models from generative tasks for dense visual prediction problems such as semantic segmentation and depth estimation. In contrast to traditional methods in dense visual prediction that generate dense outputs in one phase, DDP conceptualizes the issue as an iterative

denoising process, systematically enhancing forecasts across several steps. DDP presents a novel methodology by conceptualizing dense prediction tasks as Conditional Diffusion denoising processes, adhering to a “noise-to-map” generative framework for prediction through the incremental elimination of noise from a random Gaussian distribution, with the image as the conditioning factor.

2) *IADB*: Iterative α -(de)Blending (IADB) [9], a simplified deterministic diffusion model that transforms Gaussian noise into structured data, such as images. This approach makes diffusion models more accessible and efficient by leveraging essential blending and deblending operations, avoiding complex concepts like Langevin dynamics or stochastic differential equations (SDEs). IADB defines a process where samples from two densities (e.g., Gaussian noise and real data) are blended using an α parameter. The model iteratively “de-blends” the data, gradually reducing noise to refine the output. Unlike stochastic diffusion models, IADB creates a deterministic path between the noise and data distributions. It maps a noisy sample through several intermediate steps, progressively approximating the target distribution. This generative construction allows them to connect any two data densities irrespective of Gaussian noise, as required in the traditional diffusion models in the reverse process.

C. Improving Domain Generalization Models for Segmentation

1) *DGInStyle*: DGInStyle [4] enhances domain generalization (DG) model performance for semantic segmentation by introducing a novel data augmentation strategy that increases model robustness to unknown domain shifts. It utilizes pretrained text-to-image diffusion models, specifically fine-tuned to generate images aligned with semantic segmentation masks. Hence, it allows the generation of synthetic datasets that mimic real-world variability. Further, the Style Swap method decouples semantic content from style to avoid overfitting the source domain’s style. The pipeline generates stylized images by swapping domain-specific style layers with those from the pre-trained model, maintaining

diverse, realistic appearances. To deal with the generation of small objects due to low-resolution latent, they proposed a two-stage Multi-Resolution Latent Fusion (MRLF). In the first stage of MRLF, they perform a low-resolution pass, which serves as a reference from which large-sized classes can be extracted. In the second High-resolution pass using controlled tiled multi-diffusion, they get refinement of small-size objects, and later, the output of this pass is combined with the large-sized component of the low-resolution forward stage and used in the denoising reverse process. Thus, MRLF ensures that details of small objects are preserved without any degradation of large objects. Controlled tiled diffusion further enhances fine-grained features. It also improves model performance on rare classes (e.g., traffic signs) by prioritizing these during training and data generation phases, mitigating class imbalance issues.

D. Proposed Algorithms

Images contain texture and structure components, and night images have high inherent data uncertainty. Thus, we rely on semantic labels for our training procedure to extract structure components from a scene effectively. We used DDP and IADB concepts to bridge the gap between the incorrect predictions and the ground truth labels. Furthermore, using the Synthetic data from DGInStyle for training helped us extract meaningful information about the night images, thereby increasing our model’s generalisability. Our IND-DSP model is a plug-and-play module, which could be trained and then used on top of the DG model to generate structurally aware semantic labels.

1) *Training-Algorithm*: We used 1000 annotated DGInStyle synthetic nighttime-generated images from the GTA dataset for training our IND-DSP. We follow the procedure from DDP to encode the input image, its corresponding DG predicted label, and ground truth semantic map. Then, inspired by the IADB technique, we perform α -blending between the encoded ground truth and predicted DG semantic label to learn the bridge between ground truth and predicted label. We used the conditional map decoder from DDP to learn the difference between

the ground truth and the predicted label, which can act as a correction module. Lastly, we add the softmax layer to model prediction logits with the softmax layer to the difference logits and pass it to the cross-entropy loss function to constitute the training of our model, as shown in Figure 3 and Algorithm 1. Thus, we are learning map-decoder in our training procedure, which can be used as a correction module to learn the difference between ground truth and predicted DG label.

2) *Sampling-Algorithm*: During the sampling phase, the inputs consist of the image, the prediction map, and the prediction logit from the DG model. Subsequently, we employ the DDP procedure to encode the prediction labels and iteratively α -de-blend (for time steps T) utilizing the learned map decoder from the training phase, which functions as a correction module. Each time step enhances our predicted label, thereby making the predicted DG label more structurally aware of the scene. Figure 4 and Algorithm 2 describe further our sampling strategy.

III. EXPERIMENTAL RESULTS

The qualitative findings in figure 5 and the quantitative performance in figure 6 demonstrate a significant enhancement in RobustNet predictions. The sampling process involves setting T, the number of iterative denoising, to 5. We choose this because the performance improvement plateaus after this time step, as indicated in Table I. The input images have a resolution of 512×512, which is subsequently downscaled to 128×128 after encoding. Additionally, the logits of RobustNet predictions possess a size of 19×128×128. The batch size was set at 16. The qualitative and quantitative results demonstrate that our method effectively identifies the overall structural prior of the urban driving scene, with the Sky class positioned at the top, and the structural awareness of the road, building, and sidewalk classes has significantly improved. Additionally, it effectively corrects instances of small area classes, including traffic lights, poles, individuals, riders, and motorcycles. While DGInStyle [4] improved the RobustNet (GTA) predictions to 12.73% mIoU.

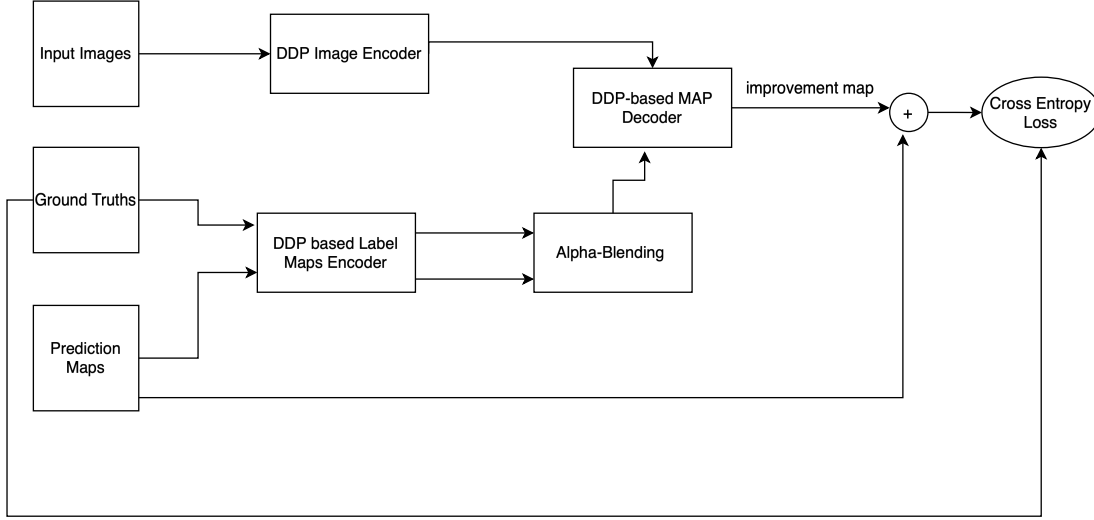


Fig. 3: IND-DSP Training Procedure

Algorithm 1 IND-DSP Training

```

def train(images, model_preds, maps, logits(model_pred), alpha):
    """images: [b, 3, h, w],
    model_pred: [b, 1, h, w],
    logits(model_pred): [b, 19, h, w],
    maps: [b, 1, h, w]"""
    img_enc = image_encoder(images)
    map_gt_enc = encoding(maps)
    map_gt_enc = (sigmoid(map_gt_enc) * 2 - 1) * scale
    map_pred_enc = encoding(model_preds)
    map_pred_enc = (sigmoid(map_pred_enc) * 2 - 1) * scale
    # alpha blending
    map_alpha = (1 - alpha) * map_pred_enc + alpha * map_gt_enc
    map_gt_diff_pred = map_decoder(map_alpha, alpha, img_enc)
    map_approx_gt = softmax(map_gt_diff_pred) + softmax(logits(model_pred).detach())
    loss = CE(map_approx_gt, maps)
    return loss

```

IV. DISCUSSION AND CONCLUSION

This research study examined stochastic diffusion models relevant to segmentation and deterministic diffusion models that facilitate translation across various data distributions. By employing DDP, IADB, and the recent data augmentation

technique DGInStyle, we devised a novel training and sampling methodology that requires minimal computational resources, obviating the necessity to retrain the DG model from the ground up, as mandated by the Naive DGInStyle approach, while also facilitating efficient sampling. It provided substantial improvements over the current state-of-the-

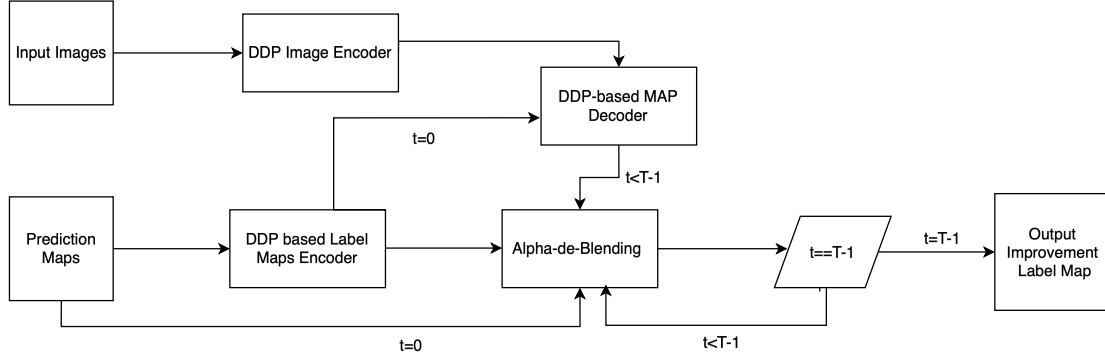


Fig. 4: IND-DSP Sampling Procedure

Algorithm 2 IND-DSP Sampling

```

def sampling(images, model_pred, logits(model_pred), T):
    """images: [b, 3, h, w],
    model_pred: [b,1, h, w]
    logits(model_pred): [b, 19, h, w]
    T: 5 (hyperparameter)"""
    map_pred = softmax(logits(model_pred))
    map_pred_enc = encoding(model_pred)
    map_pred_enc = (sigmoid(map_pred_enc) * 2 - 1) * scale
    for t in range(T):
        alpha_t = t / T
        alpha_t_plus_one = (t + 1)/T
        # alpha de-blending
        map_pred = map_pred + (alpha_t_plus_one - alpha_t) * softmax(map_decoder(images,
            map_pred_enc, alpha_t))
        map_pred = softmax(map_pred)
        map_pred_enc = encoding(argmax(map_pred))
        map_pred_enc = (sigmoid(map_pred_enc) * 2 - 1) * scale
    return map_pred

```

TABLE I: Ablation of Reverse denoising time steps

Time-Step	mIoU
1	16.53%
2	16.40%
3	17.53%
4	18.05%
5	18.16%
6	18.16%
7	18.17%

art. The subsequent study intends to investigate the proposed training and sampling methodologies across more source domain datasets and various domain-generalizable models.

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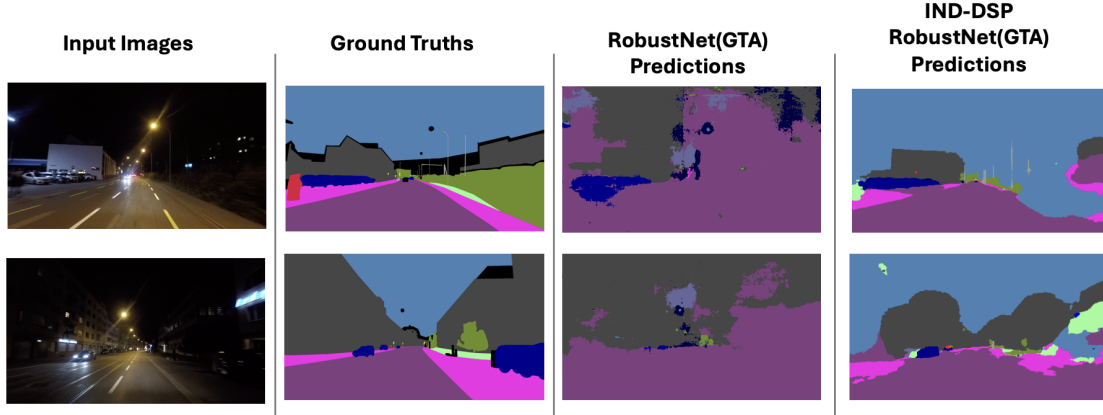


Fig. 5: IND-DSP uses the Structure prior to enhance the performance of RobustNet [1] predictions

Seg Correction: RobustNet(GTA) predicted DarkZurich VAL		
Class	Original(RobustNet) MIoU%	IND-DSP- RobustNet(Ours) MIoU%
0,Road	32.3	65.3
1,Sidewalk	2.5	23.08
2,Building	36.1	54.08
3,wall	11.4	1.83
4,fence	0.3	0.0
5,Pole	0.3	19.22
6,Traffic light	6.7	17.72
7,Traffic sign	0.0	3.79
8,Vegetation	11.1	6.87
9,Terrain	2.1	0.8
10,Sky	0.0	50.21
11,Person	3.6	21.56
12,Rider	0.0	19.96
13,Car	25.4	48.92
14,Truck	0.0	0.0
15,Bus	0.0	0.0
16,Train	0.0	0.0
17,Motorcycle	0.9	11.74
18,Bicycle	0.0	0.0
MIoU	7.3%	18.16%

Fig. 6: Quantitative performance of IND-DSP to improve the performance of RobustNet [1] predictions

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